

SDG 3: Good Health and Well-Being – Nutrition Tracking System

Proponents:

Monderondo, Joseph Zalde B.

Trigo, Karen S.

Valencia, Noriel

Program:

Course: IT Electives I

(Final Project)

Instructor:

Neypes, Justin Louise R.

Date Submitted: May 20, 2026.

Table of Contents

INTRODUCTION.....	
SYSTEM ANALYSIS.....	
SYSTEM DESIGN.....	
SYSTEM IMPLEMENTATION.....	
TESTING AND EVALUATION.....	
SYSTEM FEATURES DESCRIPTION.....	
INDIVIDUAL CONTRIBUTIONS.....	
CONCLUSIONS.....	
REFERENCE.....	
APPENDICES.....	

I. INTRODUCTION

Background of the Study

A **Food Calorie Tracker** designed for people with health complications, such as high blood pressure, diabetes, those on strict diets, and elderly individuals, goes beyond basic calorie counting. Its main purpose is to help users monitor, manage, and improve their daily food intake in a safe and guided way.

At its core, the system allows users to log meals and track calories, but it also includes more health-focused features. For individuals with conditions like diabetes or hypertension. This helps users make better food choices that align with medical recommendations.

The target users often need simple, clear, and accessible design, especially elderly users. So the interface should use large text, easy navigation, and minimal steps when logging food. For those on a diet or with medical restrictions.

In summary, this type of calorie tracker is not just a fitness tool, it is a health management system designed to support vulnerable users by promoting safe eating habits, preventing complications, and improving overall well-being.

Problem Statement

Many individuals with health complications such as diabetes, high blood pressure, obesity, and elderly-related conditions struggle to manage their daily food intake properly. Existing calorie tracking applications are often designed to track their nutritional intake for everyday use such as sugar, sodium, and calorie intake warnings. Additionally, elderly users may experience difficulty using complicated interfaces due to small text, confusing navigation, and complex food logging processes.

As a result, users may unintentionally consume unhealthy foods that can worsen their medical conditions. There is a need for a Food Calorie Tracker System that provides accurate monitoring, simple accessibility, and health-based guidance to help users maintain safe and balanced eating habits.

Objective of the System

General Objective

To develop a Food Calorie Tracker System that helps users with health complications monitor and manage their daily food intake effectively and safely.

Specific Objective

1. To allow users to record and monitor their daily meals and calorie intake.
2. To provide nutritional information such as sugar, sodium, fat, and carbohydrate content.
3. To help users make healthier food choices based on their medical conditions.
4. To design a simple and user-friendly interface suitable for elderly users.
5. To generate food intake summaries and health monitoring reports.
6. To promote healthy eating habits and reduce health risks caused by poor diet management.

Scope and Limitations

Scope

The study focuses on the development of a Food Calorie Tracker System intended for individuals with diabetes, high blood pressure, strict dietary needs, and elderly users. The system allows users to:

- Register and manage personal accounts
- Log daily food intake
- Track calories and nutritional values
- Receive food intake reminders and warnings
- View daily and weekly food summaries
- Access a simple and accessible user interface

The system may be accessible through mobile or web platforms for easier monitoring and convenience.

Limitations

- The system only provides food monitoring assistance and does not replace professional medical advice.
- Nutritional information depends on the accuracy of the food database entered into the system.
- The system requires internet access for online features and database synchronization.
- The application focuses mainly on food tracking and does not include full medical diagnosis functionalities.

- Some advanced features such as AI-based meal recommendations may not be included in the initial version.

Significance of the Study

This study is beneficial to the following:

1. **Users with Health Complications**

The system helps users monitor their food intake and maintain healthier eating habits, reducing risks related to diabetes, hypertension, and obesity.

2. **Elderly Individuals**

The simple and accessible interface allows elderly users to easily use the system without difficulty.

3. **Healthcare Providers**

Doctors and nutritionists may use the system's reports to monitor patients' dietary habits and provide better recommendations.

4. **Researchers and Future Developers**

The study can serve as a reference for future research and improvements in health monitoring and nutrition management systems.

5. **Schools and Institutions**

The study contributes knowledge in the fields of Information Technology, Health Informatics, and System Development.

II. SYSTEM ANALYSIS

Existing System

Existing calorie tracking applications focus on food nutritions, fitness and weight loss. Most systems provide basic calorie counting features also for users with medical conditions such as diabetes and hypertension etc. Some applications also have complex interfaces that may be difficult for elderly individuals to use.

Weaknesses of Existing Systems

- Limited health-specific monitoring.

- Complicated navigation for elderly users.
- Lack of warning notifications for unhealthy food intake.

Proposed System

The proposed Food Calorie Tracker System is designed specifically for users with health complications. The system not only tracks calories but also monitors nutritional values such as sugar, sodium, carbohydrates, and fat intake etc.

The proposed system includes:

- Simple and user-friendly interface
- Health-based food monitoring
- Food warnings and notifications
- Daily meal tracking and summaries
- Accessible design for elderly users

This system aims to provide a safer and more guided approach to food management.

Functional Requirements

1. User Registration and Login
2. Food Logging and Meal Recording
3. Calorie and Nutritional Information Tracking
4. Daily and Weekly Reports Generation
5. Food Intake Notifications and Warnings
6. User Profile Management
7. Search Food Database Feature
8. Health Restriction Monitoring
9. Dashboard for Food Intake Summary

Non-Functional Requirements

- **Usability**

The system must be easy to use with large text, simple buttons, and easy navigation.

- **Reliability**
The system should provide accurate calorie and nutritional information consistently.
- **Performance**
The application should respond quickly during food logging and report generation.
- **Security**
User data and health information must be protected and secured.
- **Accessibility**
The system should support elderly users and users with limited technical knowledge.
- **Maintainability**
The system should allow future updates and feature improvements easily.

III. SYSTEM DESIGN

System Architecture

The proposed system follows a three-tier architecture:

1. Presentation Layer

This layer contains the user interface where users interact with the system using mobile or web applications (VB.NET window forms)

2. Application Layer

This layer processes user requests, computes calorie intake, validates food data, and manages notifications and reports (Validation and Processing)

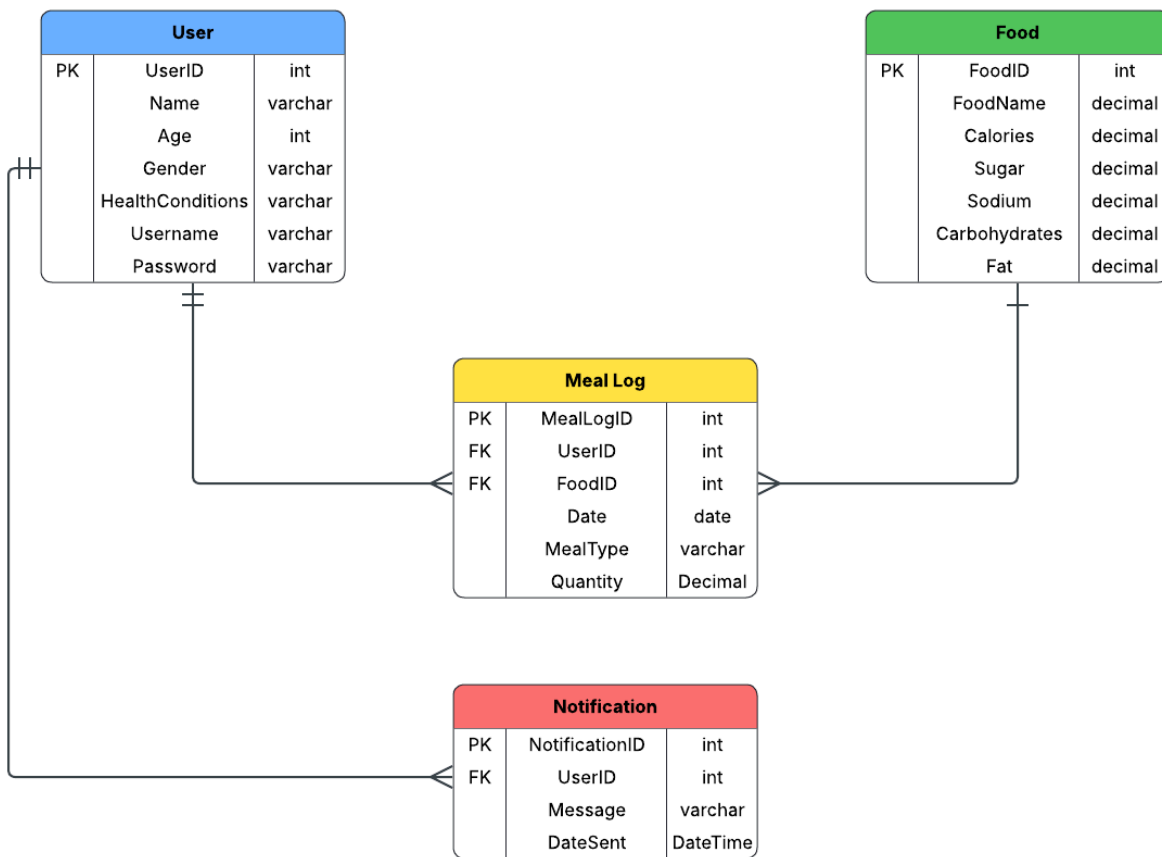
3. Database Layer

This layer stores user information, food databases, meal records, nutritional values, and reports (SQL Server LocalDB using ADO.NET)

Flow of the System

- IV. User logs into the application.
- V. Users record food intake.
- VI. The system processes nutritional information.
- VII. The database stores the meal records.
- VIII. The system displays calorie summaries and warnings.

Entity Relationship Description



Entities and Their Description

1. User

Stores personal information of users.

- UserID (Primary Key)
- Name
- Age
- Gender
- HealthCondition
- Username
- Password

2. Food

Contains nutritional details of food items.

- FoodID (Primary Key)
- FoodName
- Calories
- Sugar
- Sodium
- Carbohydrates
- Fat

3. MealLog

Records the food consumed by users.

- MealLogID (Primary Key)
- UserID (Foreign Key)
- FoodID (Foreign Key)
- Date
- MealType
- Quantity

4. Notification

Stores warning and reminder messages.

- NotificationID (Primary Key)
- UserID (Foreign Key)
- Message
- DateSent

Relationship Description

- One User can have many Meal Logs.
- One Food item can appear in many Meal Logs.
- One User can receive many Notifications.

- MealLog connects User and Food entities through foreign keys.

IX. SYSTEM IMPLEMENTATION

Programming language and tools

Tools

- **Label**
- **TextBox**
- **Button**
- **Data Grid View**

Database Connectivity

The system uses ADO.NET SqlClient to establish communication between the VB.NET Windows Forms Application and SQL Server LocalDB. The database connection is handled inside the HealthLogic class which centralizes reusable database functions and logic operations.

CRUD Operations

Create

Allow users to:

1. Log in Daily Meals and Nutrition
2. View Ongoing Food Diary
3. Calculate Total Nutrition Intake
4. Receive Automatic Health Alerts

Read

Allow users to display the history

1. Populating the data grid
2. The Summary Calculation

Update

Allow to save the changes:

1. Track which item is being edited
2. Load the selected meal into the text boxes
3. Handle the actual "Update" in your Save button

Delete

allow to remove:

1. Delete Meal

Validation Features

- Mandatory Text Validation
- Numeric validation
- All-or-Nothing Conditional Logic

X. TESTING AND EVALUATION

- Database connectivity
- CRUD functionality
- User interaction

XI. SYSTEM FEATURES DESCRIPTION

User Profile Management

The system allows users to enter and manage their personal information such as name, age, gender, and health condition. This helps the application provide proper food monitoring based on the user's health needs.

Features:

- User information management
- Health condition recording
- User profile storage
- Editable personal information

Meal Logging

The Meal Logging feature allows users to record the food they consume daily. Users can input food names and nutritional information such as calories, sugar, sodium, carbohydrates, and fat.

Features:

- Daily meal recording
- Nutritional information input
- Food consumption tracking
- Data storage in SQL Server database

Calorie and Nutrition Monitoring

The system automatically calculates and monitors the user's daily calorie intake together with sugar and sodium levels. This helps users avoid unhealthy food consumption.

Features:

- Calorie monitoring
- Sugar intake tracking
- Sodium intake tracking
- Nutritional summary computation

Dashboard

The Dashboard serves as the main interface of the application where users can view summaries of their daily nutritional intake and access different system functions.

Features:

- Daily intake summary

- Quick access navigation
- Health status display
- User-friendly interface

Progress Monitoring

The Progress Monitoring feature visually displays the user's daily nutritional progress using progress bars. It helps users easily understand whether they are within the recommended intake limits.

Features:

- Progress bar indicators
- Daily nutritional monitoring
- Intake limit visualization
- Real-time updates

Reminder and Alerts

The system provides reminder notifications and health alerts to encourage users to maintain healthy habits and avoid excessive nutrient intake.

Features:

- Health reminder notifications
- Warning alerts for high sugar or sodium intake
- Daily health reminders
- User alert messages

Reports and Meal Summary

The Reports feature allows users to view stored meal records and nutritional summaries using a DataGridView table.

Features:

- Meal history viewing
- Nutritional report generation

- DataGridView display
- Record retrieval from database

Database Management

The system uses SQL Server database management for storing and retrieving user meal logs and nutritional information securely.

Features:

- SQL database integration
- Secure data storage
- CRUD operations
- Data retrieval and management

User-Friendly Interface

The system interface is designed to be simple, clear, and accessible, especially for elderly users and individuals with limited technical knowledge.

Features:

- Large buttons and labels
- Simple navigation
- Readable fonts
- Organized interface layout

XII. CONCLUSIONS

The Food Calorie Tracker System successfully provides a reliable and user-friendly solution for monitoring daily food intake and nutritional health. The system helps individuals with diabetes, hypertension, dietary restrictions, and elderly users maintain healthier eating habits through calorie and nutrient tracking.

The study demonstrated that integrating health monitoring with a simple interface improves usability and encourages better food management practices. The system also provides accurate data storage, reporting, and reminder functionalities that support users in making healthier lifestyle decisions.

Overall, the proposed system contributes to health awareness, preventive care, and improved well-being through technology-based dietary monitoring.

XIII. REFERENCE

- Amugongo, L. M., Kriebitz, A., Boch, A., & Lütge, C. (2022). Mobile Computer Vision-Based Applications for Food Recognition and Volume and Calorific Estimation: A Systematic Review. *Healthcare*, 11(1), 59. <https://doi.org/10.3390/healthcare11010059>
- Burrows, T. L., & Rollo, M. E. (2019). Advancement in Dietary Assessment and Self-Monitoring using technology. *Nutrients*, 11(7), 1648. <https://doi.org/10.3390/nu11071648>
- Ramkumar, V. (2022). Android based monitoring system with diet and calorie tracker. *IJERT*. <https://doi.org/10.17577/IJERTCONV10IS09028>
- Dugas, K., Giroux, M., Guerroudj, A., Leger, J., Rouhafzay, A., Rouhafzay, G., & Jbilou, J. (2025). Calorie-Counting Apps for Monitoring and Managing Calorie Intake in Adults Living with Weight-Related Chronic Diseases: Decade-Long Scoping Review (2013-2024). *JMIR Mhealth and Uhealth*, 14, e64139. <https://doi.org/10.2196/64139>

XIV. APPENDICES

Appendix A – System Interface

Contains screenshots of:

- Login Form
- Dashboard
- Meal Logging Form
- Reports Section
- Reminder Notifications

Appendix B – Database Files

The system uses SQL Server database files:

- FoodCalorieTracker.mdf
- FoodCalorieTracker.ldf

These database files are connected to the VB.NET Windows Forms Application using ADO.NET SqlConnection to save and retrieve meal records, calorie information, and nutrition data.

Appendix C – Source Code

Includes:

- SQL Connection Code
- CRUD Operations
- DataGridView Functions
- Progress Bar Functions
- Reminder Functions

Appendix D – Sustainable Development Goal Alignment

The project supports:

SDG 3 – Good Health

The Food Calorie Tracker System helps users monitor and manage their daily nutritional intake digitally. The system promotes healthy eating habits, calorie monitoring, and nutrition awareness for individuals with diabetes, hypertension, strict diets, and elderly users.

The application supports preventive healthcare by encouraging proper food management and reducing health risks associated with unhealthy eating habits.